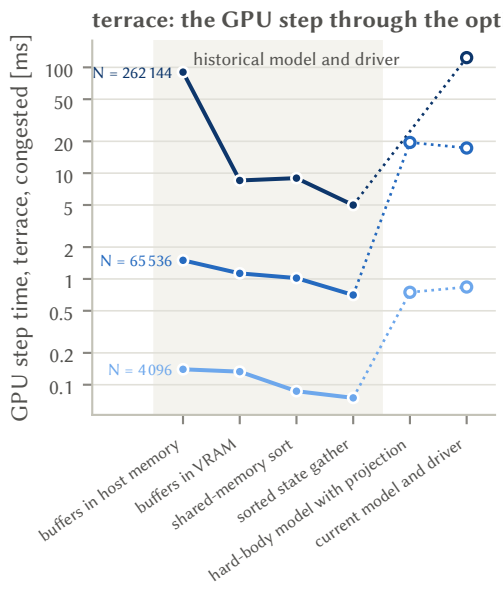
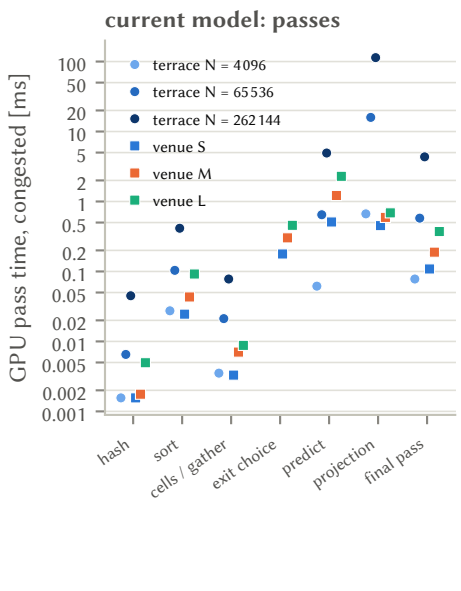

terrace: the GPU step through the optimisations
historical model and driver
GPU step time, terrace, congested [ms]
100
50
20
10
5
2
1
0.5
0.2
0.1
N = 262 144
N = 65 536
N = 4 096
buffers in host memory
buffers in VRAM
shared-memory sort
sorted state gather
hard-body model with projection
current model and driver


current model: passes
GPU pass time, congested [ms]
100
50
20
10
5
2
1
0.5
0.2
0.1
0.05
0.02
0.01
0.005
0.002
0.001
terrace N = 4 096
terrace N = 65 536
terrace N = 262 144
venue S
venue M
venue L
hash
sort
cells / gather
exit choice
predict
projection
final pass